GROUP PROJECT

DESCRIPTIVE ANALYSIS

UNDERSTANDING THE CONCEPT

PRESENTED BY
L056 YASH SHAH
L076 SANJEET TUTEJA
L068 UMESH TIWARI

IBSAM

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INTRODUCTION
TO DESCRIPTIVE STATISTICS

Descriptive statistics summarize and understand data distribution, central tendency, and variability

KEY CONCEPTS

1. **Measures of Central Tendency**
   a. Mean -Average of data points
   b. Median: Middle value when data points are sorted
      Odd count: Median = Middle value
      Even count: Median = Average of two middle values
   c. Mode: Most frequently occurring value

$$\text{Mean} = \frac{\sum X}{N}$$

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KEY CONCEPTS

1. **Measures of Variability (Dispersion)**
   a. Range: Difference between highest and lowest valueas

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

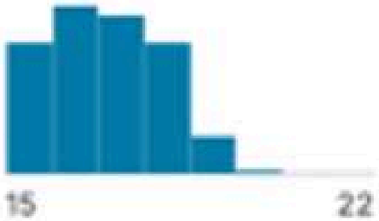
$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$

VARIANCE

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}},$$

STANDARD DEVIATION

2. **Visualizing Data**
   a. Histograms for distribution patterns
   b. Box Plots for identifying outliers and quartile ranges
   c. Bar Charts for categorical data comparisons

sex student's sex	age student's age	parent_status parent's cohabitation status	school_choice_re... reason to choose this school	class_failures number of past class failures	school_support extra educational support
F 53% M 47%		Living together 90% Apart 10%	course 37% home 28% Other (141) 36%		true 51 13% false 344 87%
F	18	Apart	course	0	yes
F	17	Living together	course	0	no
F	15	Living together	other	3	yes

family_support family educational support	higher_ed wants to pursue higher education	internet_access Internet access at home	family_relationship quality of family relationships	weekday_alcohol workday alcohol consumption	weekend_alcohol weekend alcohol consumption	health current health status	absences number of school absences
true 242 61% false 153 39%	true 375 95% false 20 5%	true 329 83% false 66 17%					
no	yes	no	4	1	1	3	6
yes	yes	yes	5	1	1	3	4
no	yes	yes	4	2	3	3	18
yes	yes	yes	3	1	1	5	2
yes	yes	no	4	1	2	5	4
yes	yes	yes	5	1	2	5	18

PROBLEM STATEMENT

IN THE CURRENT EDUCATIONAL LANDSCAPE, UNDERSTANDING THE FACTORS THAT INFLUENCE STUDENT PERFORMANCE IS CRUCIAL. THIS STUDY SEEKS TO EXPLORE THE RELATIONSHIP BETWEEN VARIOUS LIFESTYLE, FAMILY, AND ACADEMIC FACTORS TO DETERMINE THEIR IMPACT ON STUDENT SUCCESS.

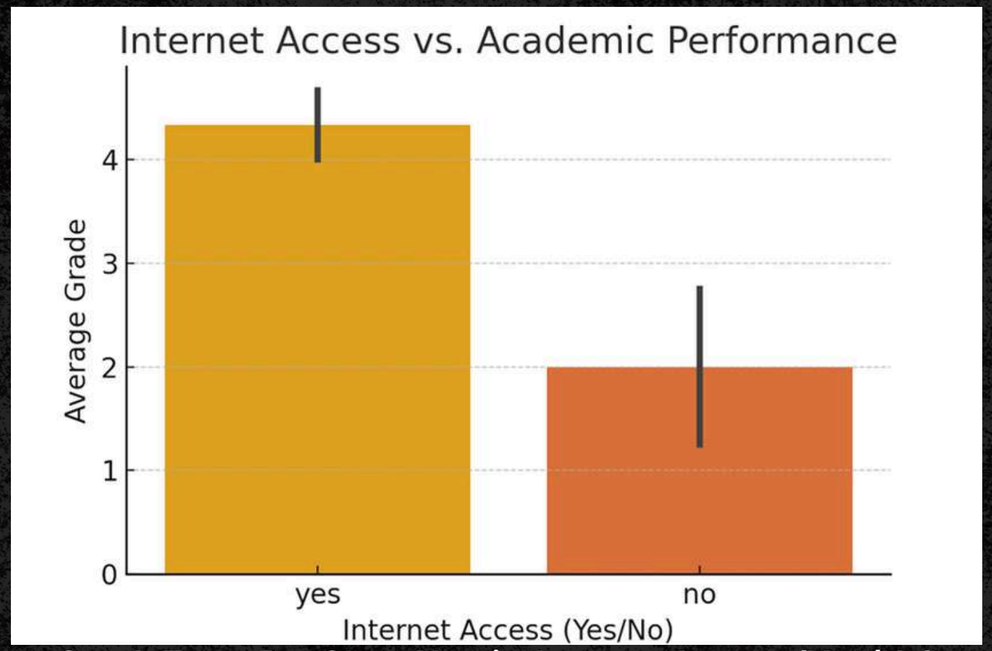
KEY QUESTIONS

- Does internet access at home significantly influence academic performance?
- How do family relationships correlate with school success?
- Is there a noticeable impact of alcohol consumption on student grades?
- What role does parental status play in shaping academic outcomes?

OBJECTIVE

By analyzing these variables through correlation and regression techniques, this research aims to identify the strongest predictors of academic performance and provide actionable insights to educators, parents, and policymakers.

INTERNET ACCESS VS. ACADEMIC PERFORMANCE



Graph Type: Bar Chart X-axis: Internet Access (Yes/No)
Y-axis: Average Grade

Explanation:
The bar chart displays the average grades of students categorized by internet access availability. Students with internet access generally show higher grades compared to those without.

Conclusion:

- Good Grades: Scores closer to 5 (indicating strong academic performance)
- Poor Grades: Scores closer to 1 (indicating weak academic performance)

This suggests that internet access can positively influence academic outcomes, potentially by aiding research, homework, and project preparation

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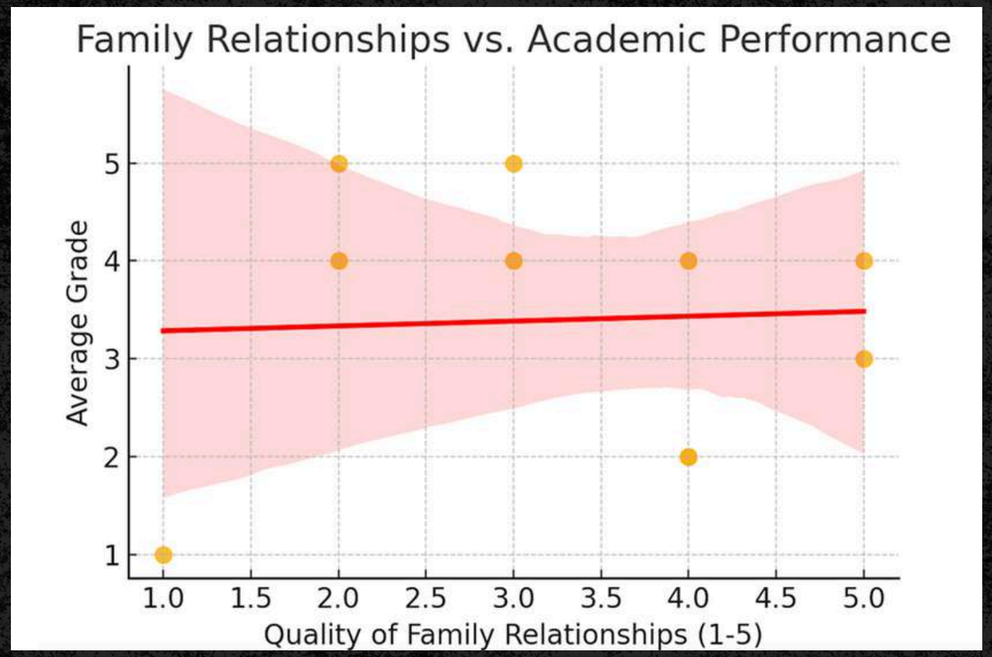
FAMILY RELATIONSHIPS VS. ACADEMIC PERFORMANCE



Graph Type: Scatter Plot with Trend Line X-axis: Quality of Family Relationships (Scale of 1-5) Y-axis: Average Grade

Explanation:

This scatter plot shows how family relationship quality correlates with student performance. The trend line suggests a positive correlation where better family relationships align with higher grades.

Conclusion:

- Good Grades: Scores closer to 5 (indicating strong academic performance)
- Poor Grades: Scores closer to 1 (indicating weak academic performance)

This highlights the importance of a supportive family environment in achieving better educational results.

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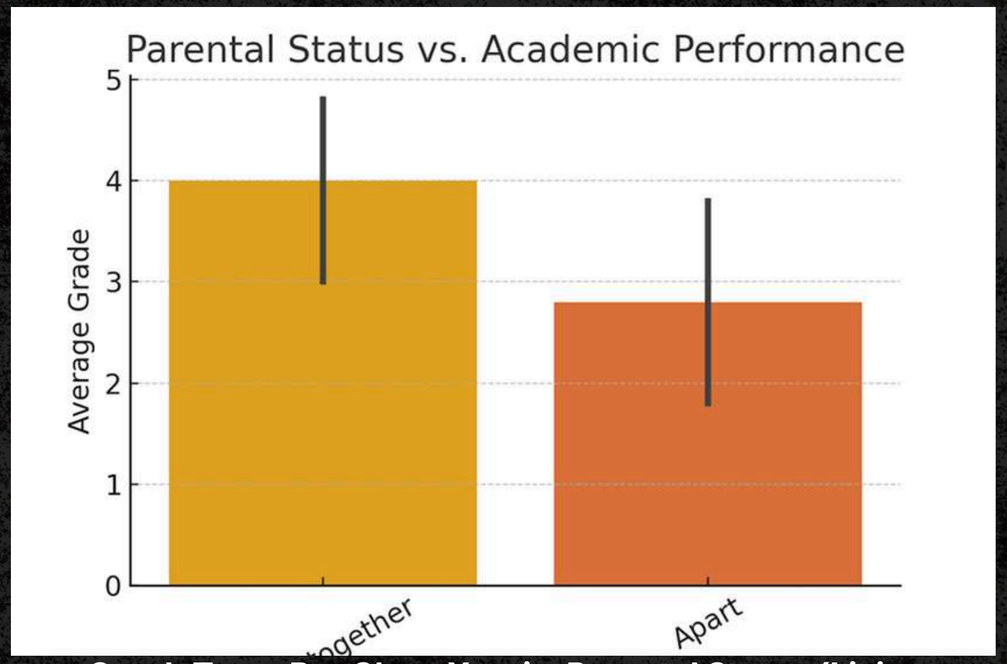
PARENTAL STATUS VS. ACADEMIC PERFORMANCE



Graph Type: Bar Chart X-axis: Parental Status (Living Together/Apart) Y-axis: Average Grade

Explanation:

The graph illustrates that students whose parents live together tend to achieve slightly better academic results compared to those from separated households.

Conclusion:

- Good Grades: Scores closer to 5
- Poor Grades: Scores closer to 1

While parental status shows some influence, other factors such as family support and school environment may also play a crucial role in student success.

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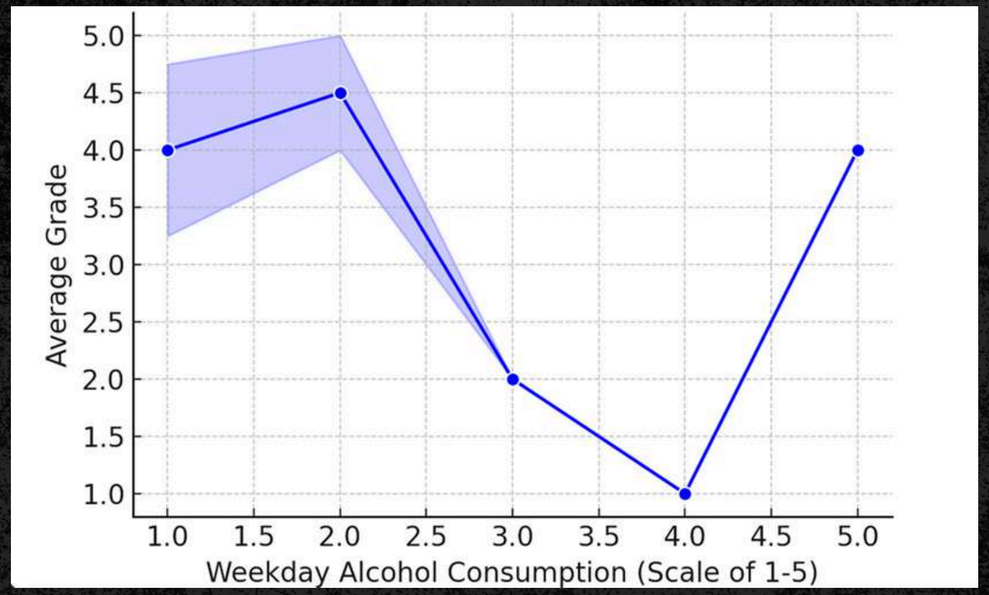
ALCOHOL CONSUMPTION VS. ACADEMIC PERFORMANCE



Graph Type: Box Plot X-axis: Weekday & Weekend Alcohol Consumption (Scale of 1-5) Y-axis: Average Grade

Explanation:

The box plot reveals the impact of alcohol consumption on academic performance. Higher alcohol consumption levels are associated with a decline in grades, particularly for frequent weekday drinking.

Conclusion:

- Good Grades: Scores closer to 5
- Poor Grades: Scores closer to 1

Students who consume less alcohol tend to have better academic outcomes, emphasizing the impact of responsible habits on education.

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PRACTICAL SOLUTIONS FOR IMPROVED ACADEMIC OUTCOMES

- STUDENTS: BY UNDERSTANDING THESE FACTORS, STUDENTS CAN ADOPT BETTER STUDY HABITS, BALANCE SOCIAL BEHAVIORS, AND UTILIZE INTERNET RESOURCES EFFECTIVELY TO IMPROVE THEIR ACADEMIC OUTCOMES.
- TEACHERS: EDUCATORS CAN USE THESE INSIGHTS TO IDENTIFY AT-RISK STUDENTS AND PROVIDE TAILORED SUPPORT TO IMPROVE PERFORMANCE.
- PARENTS: RECOGNIZING THE ROLE OF FAMILY STABILITY, INTERNET ACCESS, AND LIFESTYLE HABITS CAN HELP PARENTS CREATE A MORE CONDUCIVE LEARNING ENVIRONMENT.
- SCHOOLS: SCHOOLS CAN INTRODUCE PROGRAMS THAT PROMOTE HEALTHY BEHAVIORS, FAMILY ENGAGEMENT, AND IMPROVE ACCESS TO LEARNING RESOURCES.
- POLICYMAKERS: BY SUPPORTING AFFORDABLE INTERNET ACCESS, FAMILY COUNSELING PROGRAMS, AND AWARENESS CAMPAIGNS, POLICYMAKERS CAN HELP IMPROVE EDUCATIONAL OUTCOMES ON A BROADER SCALE.

CONCLUSION

- Data analysis highlighted key factors influencing student academic performance.
- Improved internet access correlates with better grades, underscoring the importance of digital resources.
- Strong family relationships play a vital role in boosting student outcomes.
- High alcohol consumption negatively affects academic performance, emphasizing the need for awareness programs.
- Students from separated households may require additional support from teachers, parents, and counselors.
- Combining data insights with targeted interventions can significantly enhance student success.

REFERENCES

- Dataset Source: https://www.kaggle.com/datasets/dillonmyrick/high-school-student-performance-and-demographics?resource=download
- Internet Access Impact Study: https://www.researchgate.net/
- Family Relationships & Student Outcomes: https://www.tandfonline.com/
- Alcohol Consumption & Cognitive Effects: https://www.ncbi.nlm.nih.gov/
- Parental Status and Educational Support: https://www.tandfonline.com/
- Digital Inequality in Education: https://www.sciencedirect.com/
- Impact of Lifestyle on Student Performance: https://link.springer.com/
- Education Support Strategies for Parents: https://eric.ed.gov/

THANK YOU

OUR LOVELY PROFESSOR

DR. KHINAL PARMAR